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Portable depilation instrument

Abstract

A portable depilation instrument includes a shell, and a depilation mechanism and a heat dissipation mechanism inside the shell. The shell is provided with at least two outlets and one inlet, the shell introduces an external cooling medium from the inlet, the heat dissipation mechanism includes a heat conductive component, the depilation mechanism includes a refrigeration element and an emitter, the heat conductive component isolates the refrigeration element from the emitter in two mutually sealed channels, namely, a first channel and a second channel, the first channel and the second channel are respectively in communication with two outlets, and one of the outlets allows the first channel or the second channel to pass through inside of the entire shell from one end to the other end. Due to complete isolation and a push force, air can be discharged only from a respective corresponding outlet after a heat exchange.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6416521	12/2001	Waldner et al.	N/A	N/A
2005/0177139	12/2004	Yamazaki et al.	N/A	N/A
2012/0165682	12/2011	Keeney et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2490631	12/2001	CN	N/A
2928026	12/2006	CN	N/A
201226633	12/2008	CN	N/A
101764510	12/2009	CN	N/A
101848627	12/2009	CN	N/A
102271478	12/2010	CN	N/A
102480901	12/2011	CN	N/A
202776542	12/2012	CN	N/A
103379794	12/2012	CN	N/A
103796487	12/2013	CN	N/A
203970544	12/2013	CN	N/A
203970545	12/2013	CN	N/A
104582405	12/2014	CN	N/A
205698924	12/2015	CN	N/A
206063571	12/2016	CN	N/A

207284010	12/2017	CN	N/A
207164447	12/2017	CN	N/A
108056816	12/2017	CN	N/A
207679871	12/2017	CN	N/A
207804371	12/2017	CN	N/A
208114942	12/2017	CN	N/A
208426208	12/2018	CN	N/A
109328487	12/2018	CN	N/A
208677571	12/2018	CN	N/A
208710053	12/2018	CN	N/A
209464502	12/2018	CN	N/A
209572304	12/2018	CN	N/A
113907872	12/2021	CN	N/A
2009508546	12/2008	JP	N/A
2021003950	12/2020	WO	N/A
2021027261	12/2020	WO	N/A

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Background/Summary

CROSS REFERENCE TO THE RELATED APPLICATIONS

(1) This application is the national phase entry of International Application No. PCT/CN2019/120457, filed on Nov. 23, 2019, which is based upon and claims priority to Chinese Patent Applications No. 201921052794.0, filed on Jul. 7, 2019, No. 201921467936.X, filed on Sep. 4, 2019, and No. 201921999189.4, filed on Nov. 18, 2019, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to the field of depilation and beauty, and specifically, to a portable depilation instrument.

BACKGROUND

(3) A light depilation technology is a non-invasive modern depilation technology, in which unwanted hair can be safely, quickly, and efficiently removed from a body. Specifically, light depilation is based on a theory of selective photothermolysis, in which a specific wavelength of light is allowed to penetrate epidermis without damaging epidermal hair follicles, and melanin in a hair shaft selectively absorbs light energy, so that the hair follicles are coagulated and necrotic during heating, to effectively slow down growth, thereby implementing depilation.

(4) However, it is well known that all light generates a lot of heat during use, which brings a burning pain to skin, especially a household portable depilation instrument. To help users hold and use, all the portable depilation instruments are designed relatively small, and therefore, it is easier to form heat accumulation. In this case, if heat dissipation and temperature reduction cannot be implemented in time, a relatively strong pain and a redness and swollen phenomenon are brought to the users, and even a security problem is caused.

(5) Therefore, in the prior art, to reduce heat generated when the portable depilation instrument performs depilation, a sealing member is disposed in a heat dissipation structure, and an opening is provided on the sealing member and is in communication with outside, to match a heat dissipation fin to independently form a channel for a heating member that needs to be cooled to perform heat dissipation, thereby effectively improving the effect of heat dissipation and temperature reduction.

However, the portable depilation instrument has more than one element that generates heat, and generally, internal circuits and other elements also generate heat. Consequently, how to make all the elements that generate heat dissipate the heat under a same power, and how to improve the cooling effect and maximize the efficiency are problems to be resolved currently.

SUMMARY

(6) To resolve a problem of incomplete temperature reduction of an existing portable depilation instrument, the present invention provides a portable depilation instrument.

(7) To resolve the foregoing technical problem, the present invention provides a technical solution as follows: A portable depilation instrument, including a shell, and a depilation mechanism and a heat dissipation mechanism inside the shell, where the shell is provided with at least two outlets and one inlet; the heat dissipation mechanism includes a heat conductive component; the inlet is separately in communication with different outlets, to form at least two channels; each of the channels is provided with an element that generates heat; and after being introduced from the inlet, under guidance of the heat conductive component, an external cooling medium separately passes through different channels and is then discharged from a corresponding outlet, and in this process, the external cooling medium brings heat of the separately passing region out of the shell.

(8) A portable depilation instrument includes a light-emitting body and the foregoing portable depilation instrument.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic structural diagram of a portable depilation instrument according to a first embodiment of the present invention;
- (2) FIG. 2 is a schematic exploded view of the portable depilation instrument according to the first embodiment of the present invention;
- (3) FIG. 3 is a first partial exploded view of a depilation apparatus in the portable depilation instrument according to the first embodiment of the present invention;
- (4) FIG. 4 is a second partial exploded view of a depilation apparatus in the portable depilation instrument according to the first embodiment of the present invention;
- (5) FIG. 5 is a first partial exploded view of a fan in the portable depilation instrument according to the first embodiment of the present invention;
- (6) FIG. 6 is an exploded view of a heat conductive component in the portable depilation instrument according to the first embodiment of the present invention;
- (7) FIG. 7 is a first partial exploded view of a depilation mechanism in the portable depilation instrument according to the first embodiment of the present invention;
- (8) FIG. 8 is a third partial exploded view of the depilation apparatus in the portable depilation instrument according to the first embodiment of the present invention;
- (9) FIG. 9 is a second partial exploded view of the depilation mechanism in the portable depilation instrument according to the first embodiment of the present invention;
- (10) FIG. 10 is a cross-sectional view of the depilation mechanism in the portable depilation instrument according to the first embodiment of the present invention;
- (11) FIG. 11 is a fourth partial exploded view of the depilation apparatus in the portable depilation instrument according to the first embodiment of the present invention;
- (12) FIG. 12 is an assembled diagram of the depilation apparatus in the portable depilation instrument according to the first embodiment of the present invention;
- (13) FIG. 13 is a cross-sectional view of the depilation apparatus in the portable depilation instrument according to the first embodiment of the present invention;
- (14) FIG. 14 is a partially omitted view of the portable depilation instrument according to the first

embodiment of the present invention;

(15) FIG. 15 is a partial exploded view of the portable depilation instrument according to the first embodiment of the present invention;

(16) FIG. 16 is a schematic structural diagram of a portable depilation instrument according to a second embodiment of the present invention;

(17) FIG. 17 is an expanded view of a part structure of the portable depilation instrument according to the second embodiment of the present invention;

(18) FIG. 18 is a schematic structural diagram of a heat conductive component in the portable depilation instrument according to the second embodiment of the present invention; and

(19) FIG. 19 is a schematic structural diagram of a depilation apparatus in the portable depilation instrument according to the second embodiment of the present invention.

(20) Description of reference numerals: **100**. portable depilation instrument; **1**. shell; **101**. upper shell; **102**. lower shell; **11**. inlet; **12**. head portion; **13**. holding portion; **14**. second outlet; **15**. first outlet; **16**. third outlet; **1601**. first groove; **1602**. second groove; **1603**. exhaust structure; **2**. depilation apparatus; **21**. heat dissipation mechanism; **211**. fan; **2111**. fan air inlet; **2112**. fan air outlet; **2113**. heat conductive element; **2114**. first heat dissipation fin; **212**. heat conductive component; **2121**. first vent; **2122**. second vent; **2123**. second heat dissipation fin; **2124**. third vent; **22**. depilation mechanism; **221**. cold compress portion; **222**. emitter; **223**. refrigeration element; **2231**. heating surface; **2232**. refrigeration surface; **224**. skin detection portion; **225**. reflecting plate; **226**. insulating plate; **3**. circuit apparatus; and **4**. power supply apparatus.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(21) To make the objectives, technical solutions, and advantages of the present invention clearer and more comprehensible, the following further describes the present invention in detail with reference to the accompanying drawings and embodiments. It should be understood that specific embodiments described herein are only intended to explain the present invention, but are not intended to limit the present invention.

Embodiment 1

(22) Referring to FIG. 1, the present invention provides a portable depilation instrument **100**, configured to remove unwanted hair from a human body, and including a shell **1** and a cold compress portion **221** having a light-transmitting and cold compress effect. A user may tightly attach the cold compress portion **221** to a human skin surface, the portable depilation instrument **100** emits light, and the light passes through the cold compress portion **221** and penetrates the skin surface to reach a root of a hair follicle, so that the hair follicle is coagulated and necrotic, to remove hair.

(23) The shell **1** mainly plays a role of protecting internal elements, and an overall shape of the shell is not limited, may be U-shaped, T-shaped, shuttle-shaped, or long strip-shaped, or is optionally long striped cylindrical.

(24) The shell **1** includes a holding portion **13** and a head portion **12**, the head portion **12** is at one end facing human skin, and the holding portion **13** is parallel with the head portion **12**. After the head portion **12** is fixedly connected to the holding portion **13**, an accommodating space is formed inside, and the cold compress portion **221** is exposed in the accommodating space from the head portion **12**, to be used as a carrier for emitting light and contacting the human skin. The holding portion **13** is used for holding the portable depilation instrument **100**, to facilitate operations of the user.

(25) Referring to FIG. 2, the shell **1** includes an upper shell **101** and a lower shell **102**, may be formed by assembling the upper shell **101** and the lower shell **102** through screw connection or clamping connection, or may be formed through integral processing, or is optionally formed through assembly in a clamping manner, to facilitate disassembly and assembly.

(26) A depilation apparatus **2**, a circuit apparatus **3**, and a power supply apparatus **4** are included inside the accommodating space of the shell **1**. The depilation apparatus **2**, the circuit apparatus **3**,

and the power supply apparatus **4** are all sequentially electrically connected. The power supply apparatus **4** may provide electric energy to the depilation apparatus **2**, so that the depilation apparatus can be used without plugging in an external circuit, or can be used by directly using an external power source, such as a dry battery or an energy storage battery. Therefore, the depilation apparatus is convenient to store and carry, and can be used outside. The circuit apparatus **3** may be externally connected to an external circuit to charge the power supply apparatus **4** and control the entire portable depilation instrument **100** on/off, power regulation, and temperature protection.

(27) A person skilled in the art may understand that, during working, all elements in the depilation apparatus **2**, the circuit apparatus **3**, and the power supply apparatus **4** generate heat.

(28) In daily production, some control elements (not shown in the figure) in the circuit apparatus **3** are generally disposed on a surface of the shell **1**, to form a switch, a screen, and an operation panel. The user may control the entire portable depilation instrument **100** through manipulation, to adjust the on/off of the portable depilation instrument **100** and control levels.

(29) The depilation apparatus **2** is disposed at one end near the head portion **12** of the shell **1**, and may be fixedly clamped inside the accommodating space of the shell **1**. An inlet **11** and a second outlet **14** corresponding to a region of the depilation apparatus **2** are disposed on one side of the shell **1**, that is, the upper shell **101** or the lower shell **102**. It may be understood that, both the inlet **11** and the second outlet **14** are disposed on a same side of the shell **1**, and are closely arranged together. This design is to save a space of the shell **1**, so that the shell **1** can be relatively shorter, and is compact and precise, and the inlet **11** and the second outlet **14** are designed together to shorten an air circulation distance of air, to implement rapid air circulation, so that the heat dissipation efficiency is higher.

(30) Further, the first outlet **15** is disposed at an end portion of the shell **1**, that is, one end of the shell **1** away from the head portion **12**. The reason why the first outlet **15** is remotely disposed is to increase a distance through which the air flows, so that a cooling range is wider.

(31) It may be understood that, during normal production, both the circuit apparatus **3** and the power supply apparatus **4** are disposed inside the shell, and are optionally disposed inside the holding portion **13**. In this embodiment, both the circuit apparatus and the power supply apparatus are disposed on a back side of the depilation apparatus **2**, that is, one side of the depilation apparatus **2** away from the human skin. By using this design, a range of the air involved is expanded, to achieve an overall cooling effect inside the shell.

(32) Referring to FIG. 3, the depilation apparatus **2** includes a heat dissipation mechanism **21** and a depilation mechanism **22**. The depilation mechanism **22** emits light to remove hair from the human body, and is mainly in contact with the human skin. The heat dissipation mechanism **21** is used for reducing a temperature, so that the human body does not have a burning sensation under the light; and the heat dissipation mechanism **21** and the depilation mechanism **22** may be connected through clamping connection or screw connection, or are optionally connected through screw connection, so that fixation is more stable.

(33) Referring to FIG. 4, the heat dissipation mechanism **21** includes a fan **211** and a heat conductive component **212**. The fan **211** may be a centrifugal, axial-flow, mixed-flow, and cross-flow fan, or is optionally a centrifugal fan, so that external air mainly flows radially after axially entering an impeller of the fan. The heat conductive component **212** and the fan **211** may be connected through screw connection or clamping connection, so that air sucked by the fan **211** may flow through the heat conductive component **212** in a split direction. The depilation mechanism **22** is connected to one surface of the heat conductive component **212** away from the fan **211**, and therefore, air flow in the heat conductive component **212** may perform cooling and temperature reduction on the depilation mechanism **22**, to achieve an ice compress effect on the human skin.

(34) Referring to FIG. 1 and FIG. 5, a specific structure of the fan **211** is as follows:

(35) The fan **211** includes a fan air inlet **2111** and a fan air outlet **2112**. A first heat dissipation fin **2114** is disposed at the fan air inlet **2111**, and the first heat dissipation fin **2114** is disposed at an end

portion of the heat conductive element **2113**. That is, the heat conductive element **2113** is located at one end near the fan air inlet **2111**, and the fan **211** is disposed at one end of the fan air inlet **2111** facing the inlet **11** on the shell **1**, to suck air through inside of the inlet **11** on the shell **1**, where the first heat dissipation fin **2114** completely covers the fan air inlet **2111**, so that before entering the fan air inlet **2111**, the external air passes through the first heat dissipation fin **2114** for heat dissipation and temperature reduction for a first time. A portion of the heat conductive element **2113** is inserted into the first heat dissipation fin **2114**, and a remaining portion extends outward.

(36) Further, the first heat dissipation fin **2114** may be a cast iron heat dissipation fin, a steel heat dissipation fin, and or an aluminum alloy heat dissipation fin. It may be understood that, because the first heat dissipation fin **2114** dissipates heat in the form of convection, a larger coverage area indicates a better heat dissipation effect.

(37) Further, the heat conductive element **2113** may be a heat conductive tube, and a material of the heat conductive element is selected as a sheet structure or a tubular structure made of a metal or an alloy such as silver, copper, or iron, or is optionally a sheet structure. Due to a unique metal heat transfer property, the heat conductive element **2113** may dissipate the heat integrally through the first heat dissipation fin **2114**.

(38) It may be understood that, in actual production, the first heat dissipation fin **2114** and the heat conductive element **2113** are often of an integrated structure, and the first heat dissipation fin **2114** and the heat conductive element **2113** are connected together. Under the action of metal heat transfer, as the fan **211** sucks air inward, the heat on the heat conductive element **2113** flows quickly, thereby implementing quick temperature reduction.

(39) Referring to FIG. 6, the heat conductive component **212** includes a first vent **2121**. The first vent **2121** is disposed at one end of the heat conductive component **212** facing the fan **211**, or the first vent **2121** is optionally disposed parallel to one end of the heat conductive component **212**, so that after entering through the first vent **2121** on one side, the external cooling medium is discharged in an opposite direction from the first vent **2121** on the other side. In addition, a second heat dissipation fin **2123** is disposed at a surface of one side of the heat conductive component **212**, and the second heat dissipation fin **2123** is optionally disposed on an upper side of the heat conductive component **212**.

(40) Referring to FIG. 7, the depilation mechanism **22** includes a cold compress portion **221**, an emitter **222**, a refrigeration element **223**, a skin detection portion **224**, a reflecting plate **225**, an insulating plate **226**, and a mounting component (not marked).

(41) Both the emitter **222** and the refrigeration element **223** are electrically connected to the circuit apparatus **3**, the cold compress portion **221** is clamped to the mounting component, the refrigeration element **223** is tightly attached to the cold compress portion **221** to cool the cold compress portion **221**, the insulating plate **226** is located between the cold compress portion **221** and the emitter **222** to prevent heat of the emitter **222** from being transferred to the cold compress portion **221**, and the reflecting plate **225** is disposed on one side of the emitter **222** away from the cold compress portion **221**, so that light emitted by the emitter **222** is concentrated on the cold compress portion **221**.

(42) Further, the cold compress portion **221** is made of a crystal material, and specifically may be made of sapphire, K9 glass, or crystal glass, provided that a transparent crystal material is met. Optionally, the cold compress portion is made of a sapphire material.

(43) It may be understood that, because the cold compress portion **221** is made of the sapphire, the cold compress portion **221** may be used as a light outlet. When the emitter **222** emits the light, because the sapphire has a relatively strong heat conductive property, the refrigeration element **223** and the cold compress portion **221** tightly attached to a surface of the emitter can efficiently generate a heat exchange, to achieve an optimal refrigeration effect. The cold compress portion **221** made of the sapphire may be round or rectangular, which is not limited herein. A surface of the cold compress portion **221** away from the emitter **222** is in contact with the human body, and a contact

surface may be an arc surface or a flat surface, and is preferably a flat surface.

(44) Further, the emitter **222** may be an IPL tube, and is located on one side of the cold compress portion **221** away from contact with the human body. The light emitted by the emitter **222** passes through the cold compress portion **221**, and is emitted into skin of a user. A light color emitted by the emitter is not limited, the light may be a colored light, a composite light, or the like, and a specific wavelength and a frequency are determined according to a use situation.

(45) Further, the skin detection portion **224** is electrically connected to the circuit apparatus **3** by using a capacitance touch detection principle. When the cold compress portion **221** is in contact with the skin, an internal preset capacitance detection apparatus detects whether a machine is actually in contact with the skin, to reduce a security problem caused by misoperations of the user.

(46) Furthermore, a shape of the reflecting plate **225** is not limited, provided that the light from the emitter **222** is concentrated in a direction of the cold compress portion **221**. The reflecting plate may be optionally U-shaped, an opening of the reflecting plate faces the cold compress portion **221**, and the emitter **222** is located at a center of the opening of the U-shaped reflecting plate **225**. In addition to concentrating the light, the reflecting plate also prevents a temperature of the emitter **222** during working from being dissipated to other places.

(47) Referring to FIG. **8**, the reflecting plate **225** is clamped to the heat conductive component **212**, and is disposed on a surface away from the first vent **2121**, so that the first vent **2121** forms a sealed channel.

(48) Referring to FIG. **8** and FIG. **9**, when being connected, the fan **211** and the heat conductive component **212** are sleeved on the first vent **2121** through the fan air outlet **2112**. Because a size a of the fan air outlet **2112** is greater than a size b of the first vent **2121**, an extra gap c forms a second vent **2122**, so that the air is divided into two directions in the first vent **2121** and the second vent **2122**.

(49) Referring to FIG. **11** and FIG. **12**, further, when the fan **211** and the heat conductive component **212** are connected, the heat conductive element **2113** on the fan **211** is tightly attached to the refrigeration element **223**, and the first vent **2121** is exposed on one side of the fan **211**, to discharge the air sucked by the fan **211**.

(50) Referring to FIG. **11** and FIG. **13**, the refrigeration element **223** is optionally a refrigeration element **223** in a semiconductor refrigeration manner, the refrigeration element **223** includes a heating surface **2231** and a refrigeration surface **2232** that are disposed opposite to each other, and the refrigeration surface **2232** of the refrigeration element **223** is tightly attached to the cold compress portion **221**, to cool the cold compress portion **221**.

(51) It may be understood that, due to the particularity of the principle of the light depilation technology, a lot of heat is generated when the light is emitted, and when the cold compress portion **221** is in contact with the skin, to prevent the light from generating a burning pain to the user, a structure for temperature reduction needs to be disposed inside a body of the portable depilation instrument **100**. Regardless of manners of temperature reduction, the heat is generated under the heat exchange, and the heating surface **2231** of the refrigeration element **223** is connected to the heat conductive element **2113**. In addition to the refrigeration element **223**, other internal elements, such as the emitter **222**, and the power supply apparatus **4** and the circuit apparatus **3** inside the shell **1** all generate a lot of heat.

(52) If the heat is not discharged in time, the refrigeration element **223** cannot achieve the refrigeration effect at maximum efficiency, thereby failing to reach a cooling temperature at which the skin is temporarily paralyzed.

(53) Referring to FIG. **13**, FIG. **14**, and FIG. **15**, according to the foregoing two directions of the first vent **2121** and the second vent **2122** of the air, one of the directions is the first vent **2121**, and the first vent **2121** is in communication with the first outlet **15** on the shell **1** to form a first channel A in which the air flows.

(54) It may be understood that, the external air is sucked from the inlet **11**. After entering the first

vent **2121** through the second vent **2122**, the external air passes through the emitter **222** to reduce a temperature of the emitter **222**, then passes through the circuit apparatus **3** and the power supply apparatus **4** inside the shell **1**, and is discharged from the first outlet **15** of the shell **1**, to form an air path of the first channel, and temperature reduction and cooling are also performed on regions through which the air passes. For example, the circuit apparatus **3** and the power supply apparatus **4** are disposed in the region through which the air passes, that is, one side of inside of the shell **1** near the first outlet **15**, to perform heat dissipation on the circuit apparatus **3** and the power supply apparatus **4** and bring the air out of the first outlet **15**.

(55) The heat conductive component **212** and the second heat dissipation fin **2123** at an upper end of the heat conductive component are integrated, an outer surface of the heat conductive component **212** and an inner wall of the shell **1** are enclosed to form a space, and a second vent **2122** is formed at the fan air inlet **2111** in an attachment manner. The second heat dissipation fin **2123** is disposed in the space and faces the second outlet **14**, so that the air exits in a direction towards the second outlet **14**, and therefore, the second vent **2122** and the second outlet **14** on the shell **1** form a second channel B.

(56) It may be understood that, the second heat dissipation fin **2123** not only has a function of heat dissipation, but also has a function of guiding the air. Based on this, the second heat dissipation fin is designed as a fin structure, so that internal air may directly exit towards the second outlet **14**. When passing through the second heat dissipation fin **2123**, the air flows at the refrigeration element **223**, to perform temperature reduction on the refrigeration element **223**. The reason why the air is directly discharged is that the refrigeration element **223** is an element that generates a maximum amount of heat inside the portable depilation instrument **100**, and therefore a path of the air in an air circulation needs to be reduced, so that the air is discharged at the first time, thereby achieving the cooling and temperature reduction effect at maximum efficiency.

(57) Based on the first channel A and the second channel B, in some embodiments, a manner of heat dissipation is not using the fan **211** for air cooling, but an external cooling medium such as water or coolant may be used, provided that the external medium can bring heat of the two channels.

(58) Based on the first channel A and the second channel B, in some embodiments, the heat dissipation mechanism **21** may not have the heat conductive element **2113** and the first heat dissipation fin **2114**, and may perform heat dissipation on the emitter **222** and the refrigeration element **223** by directly using air of the fan **211** or another medium.

(59) The second outlet **14** is disposed on one side of the inlet **11**, both are disposed on a side surface of the shell **1** near one end of the heat conductive element **2113**, and the first outlet **15** is located at an end portion of the shell **1** away from the heat conductive element **2113**, so that air of the first channel is able to completely pass through inside of the entire shell **1**, and an air inlet direction of the inlet **11** is parallel to an air outlet direction of the second outlet **14**, and is perpendicular to an air outlet direction of the first outlet **15**.

(60) A contact end surface between the cold compress portion **221** and the skin is disposed on a same side as an end surface at which the skin detection portion **224** is exposed, so that when light depilation is performed on the skin, the cold compress portion **221** is in contact with a periphery of the irradiated skin, to perform cold compress and temperature reduction on the periphery of the irradiated skin, to reduce a burning sensation generated by the irradiated skin. In addition, the cold compress portion **221** may be close to zero temperature, to really make the skin near the light outlet reach infinitely close to a freezing point, so that the burning sensation of the skin can be alleviated, and short-time contact does not cause skin damage.

(61) Under the guidance of the second heat dissipation fin **2123**, air entering the second vent **2122** is directly discharged from the first outlet **15** after passing through the heat conductive element **2113**. In this way, an air path of the second channel is formed, and similarly, temperature reduction and cooling are performed on the regions through which the air passes.

(62) In the present invention, an improvement is made on a portable depilation instrument using an original heat dissipation mode. The fan **211** is used to suck external air, through the heat conductive component **212**, an air direction separately flows to the first channel and the second channel, the first channel and the second channel are completely isolated, the refrigeration element **223** and the emitter **222** are respectively located in positions through which the first channel and the second channel pass, and as the air flows in the completely isolated first channel and second channel, heat dissipation is separately performed on the refrigeration element and the emitter; and the first channel may further perform temperature reduction on other heat generation elements inside the shell **1**, for example, elements such as the circuit apparatus **3** and the power supply apparatus **4**, to achieve an efficient and complete temperature reduction and cooling effect.

(63) In some embodiments, a plurality of channels more than the two channels in this embodiment are provided. Except the refrigeration element **223** and the emitter **222**, a plurality of channels may be alternatively provided, and other heat generation elements are alternatively disposed in remaining channels. This falls within the protection scope of the present invention.

(64) In some embodiments, a portable depilation instrument is provided with a light-emitting body (not shown in the figure), and structural features described in all the foregoing embodiments are used.

Embodiment 2

(65) To better resolve the foregoing problem of incomplete heat dissipation, the present invention further provides a second embodiment. The second embodiment is an extension on the basis of the portable depilation instrument **100** in the first embodiment, and includes all features of the portable depilation instrument **100** in the first embodiment.

(66) Referring to FIG. **16** and FIG. **17**, the portable depilation instrument **100** provided in the second embodiment is provided with a third outlet **16** on the shell **1**, and the third outlet **16** is optionally disposed on a peripheral side of the first vent **2121**, and is in communication with a cavity inside the shell **1**, so that the external cooling medium from the first vent **2121** may be discharged from the third outlet **16** on the peripheral side of the shell **1**.

(67) Further, in the present invention, optionally, the upper shell **101** and the lower shell **102** are respectively disposed on a first groove **1601** and a second groove **1602**, an exhaust structure **1603** capable of being in communication with the inside of the shell **1** is disposed between the first groove **1601** and the second groove **1602**, and after the upper shell **101** and the lower shell **102** are connected, a gap is formed between the first groove **1601** and the second groove **1602**, and the exhaust structure **1603** is disposed in the gap to cause the cavity inside the shell **1** to be in communication with outside, to form a third outlet **16**.

(68) Referring to FIG. **18** and FIG. **19**, furthermore, the heat conductive component **212** is provided with a third vent **2124** at the second heat dissipation fin **2123**, the third vent **2124** faces a guiding direction of the second heat dissipation fin **2123**, that is, faces a direction of the second outlet **14**, and the third vent **2124** is in communication with the first vent **2121**, so that the external cooling medium is discharged through the third vent **2124** in a direction from the second heat dissipation fin **2123** to the second outlet **14** after passing through the first vent **2121**, to better speed up a progress of the heat exchange, thereby achieving a better cooling effect on the depilation mechanism **2**.

(69) The third vent **2124** is only a means of implementing the communication of the first vent **2121**, and a quantity and a communication manner are not limited. That is, a fourth or fifth vent may be alternatively provided, or the heat conductive component **212** is correspondingly provided with at least three through holes provided on a surface of the second heat dissipation fin **2123**, so that the vent is in communication with the first vent **2121**, provided that the external cooling medium that flows inside the first vent **2121** is discharged through the second heat dissipation fin **2123**.

(70) Therefore, through the first vent **2121** and the second vent **2122**, and under the action of the

first outlet, the second outlet, the third outlet (not shown in the figure), the first vent **21**, the second vent **2122**, and the third vent **2124**, the cooling efficiency is further improved, so that better ice feeling experience is brought to the user.

(71) As summarized in Embodiment 2, the outlets disposed on the shell **1** and the vents on the depilation mechanism **2** are mainly described in this embodiment. Both a quantity of the outlets and a quantity of the vents are not limited. For example, the third outlet **16** is mainly used for discharging the external cooling medium passing through the first vent **2121** to the outside of the shell **1**, or the third vent **2124** is mainly used for discharging the external cooling medium passing through the first vent **2121** to the outside of the shell **1** through the second heat dissipation fin **2123**.

(72) A person skilled in the art should also understand that if all or a part of the components of the structure in the present invention are combined in a manner of fusion, physical connection, or the like, for example, positions of the components in the depilation apparatus **2** are moved; or the components are integrated; or the components are detachably designed; and a replacement of a feature quantity and a change in feature morphology that is not used as a function are made, all combined components may form a device/apparatus with a specific function, and such a device/apparatus replaces a corresponding component in the present invention, and also falls within the protection scope of the present invention.

(73) Compared with the prior art, the portable depilation instrument provided in the present invention has the following advantages. To help users hold, all the household portable depilation instruments are designed relatively small. Therefore, due to accumulation of a large amount of heat, heat dissipation is a very important problem, and if the heat dissipation is not good, a pain may be brought to the users during depilation, and even a security problem is caused. Previously, only a single inlet and a single outlet are provided in the portable depilation instrument, to form a channel for heat dissipation. However, it is not only a single element that generates heat inside the portable depilation instrument. However, if a plurality of heat generation elements rely on a single channel formed by one inlet and one outlet for heat dissipation, heat generated by the heat generation elements affect each other, resulting in a poor heat dissipation effect, and a working environment of the heat generation elements in the channel is affected. For example, the refrigeration element and the emitter that both generate heat are placed in a same heat dissipation channel. When passing through one of the refrigeration element and the emitter, the external cooling medium brings the heat to another place. Therefore, in the present invention, a plurality of heat dissipation channels are provided in the portable depilation instrument, and different heat generation elements may be disposed in different heat dissipation channels. The heat dissipation channel may be one inlet corresponding to a plurality of outlets, or may be a plurality of inlets corresponding to one outlet. Therefore, under the action of the heat conductive component, the external cooling medium may cool a plurality of heat generation elements inside the shell of the portable depilation instrument without interfering with each other.

(74) The foregoing descriptions are only preferred embodiments of the present invention, but are not intended to limit the present invention. Any modification, equivalent replacement, or improvement made without departing from the principle of the present invention shall fall within the protection scope of the present disclosure.

Claims

1. A portable depilation instrument, comprising: a shell, and a depilation mechanism and a heat dissipation mechanism inside the shell, wherein the shell is provided with at least two outlets and one inlet; the heat dissipation mechanism comprises a heat conductive component; and the one inlet is separately in communication with different outlets of the at least two outlets, to form at least two channels; each of the at least two channels is provided with an element generating heat; and the

heat conductive component is configured to guide an external cooling medium introduced from the one inlet to separately pass through different channels of the at least two channels and the external cooling medium is then discharged from a corresponding outlet of the at least two outlets, allowing heat of the at least two channels to be discharged out of the shell; wherein the external cooling medium is air, a fan is provided inside the shell, the fan is provided with a fan air inlet, and the fan is disposed such that one side of the fan air inlet of the fan faces the one inlet on the shell, to suck the air into an inside of the shell; and the heat dissipation mechanism further comprises a heat conductive element, a first portion of the heat conductive element is disposed between the fan air inlet of the fan and the one inlet, the air is introduced from the one inlet on the shell, the first portion of the heat conductive element is provided with a first heat dissipation fin, and a second portion of the heat conductive element extends to a refrigeration element.

2. The portable depilation instrument according to claim 1, wherein the at least two outlets comprise a first outlet and a second outlet, an outer surface of the heat conductive component is provided with a second heat dissipation fin, and the heat conductive component is provided with a first vent and a second vent; the one inlet, the first vent, and the first outlet are sequentially in communication with each other to form a first channel of the at least two channels; and the one inlet, the second vent, and the second outlet are sequentially in communication with each other to form a second channel of the at least two channels.

3. The portable depilation instrument according to claim 2, wherein the at least two outlets further comprise a third outlet, and the third outlet is in communication with an internal cavity of the shell.

4. The portable depilation instrument according to claim 3, wherein the third outlet is disposed on a side surface or one side facing away from the second outlet of the shell.

5. The portable depilation instrument according to claim 4, wherein the shell comprises an upper shell and a lower shell, the upper shell and the lower shell are respectively provided with a first groove and a second groove, an exhaust structure configured to be in communication with an inside of the shell is disposed between the first groove and the second groove, and after the upper shell and the lower shell are connected, a gap is formed between the first groove and the second groove, and the exhaust structure is disposed in the gap to cause the internal cavity of the shell to be in communication with an outside, to form the third outlet.

6. The portable depilation instrument according to claim 2, wherein the heat conductive component is provided with a third vent at the second heat dissipation fin, the third vent faces a guiding direction of the second heat dissipation fin, and the third vent is in communication with the first vent.

7. The portable depilation instrument according to claim 6, wherein the third vent is disposed between the first vent and the second outlet.

8. The portable depilation instrument according to claim 2, wherein the external cooling medium is air, the depilation mechanism comprises a refrigeration element and an emitter, the refrigeration element is located in a first region, wherein the second channel passes through the first region, and the emitter is located in a second region, wherein the first channel passes through the second region.

9. The portable depilation instrument according to claim 1, wherein the second outlet is disposed on one side of the one inlet, the second outlet and the one inlet are disposed on a side surface of the shell near one end of the heat conductive element, and the first outlet is located at an end portion of the shell away from the heat conductive element, so that air of the first channel completely passes through an inside of the shell, and simultaneously takes the heat inside the shell out, and an air inlet direction of the one inlet is parallel to an air outlet direction of the second outlet, and the air inlet direction of the one inlet is perpendicular to an air outlet direction of the first outlet.

10. The portable depilation instrument according to claim 1, wherein the depilation mechanism further comprises a cold compress portion, and the shell comprises a holding portion and a head portion, wherein the head portion is provided with an opening throughout the head portion, and a

portion of the cold compress portion passes through the opening, and the portion of the cold compress portion is exposed on a surface of the shell, to be in contact with a human skin.

11. The portable depilation instrument according to claim 10, wherein the cold compress portion is made of a crystal material, and the emitter is also disposed inside the opening, and the emitter is located on one side of the cold compress portion away from contact with a human body, so that a light emitted by the emitter passes through the cold compress portion, to be emitted into a skin of a user.

12. The portable depilation instrument according to claim 10, further comprising: a circuit apparatus and a power supply apparatus, wherein the circuit apparatus and the power supply apparatus are located inside the shell, and the circuit apparatus and the power supply apparatus are located in the second region, the circuit apparatus and the power supply apparatus are electrically connected to the refrigeration element, the refrigeration element is a semiconductor refrigeration element, the refrigeration element comprises a refrigeration surface and a heating surface, wherein the refrigeration surface and the heating surface are disposed opposite to each other, and the refrigeration surface of the refrigeration element is tightly attached to the cold compress portion, to cool the cold compress portion; and the heating surface of the refrigeration element is connected to a heat conductive element, to dissipate the heat and also cool the cold compress portion.

13. The portable depilation instrument according to claim 12, further comprising: a skin detection part, electrically connected to the circuit apparatus.

14. The portable depilation instrument according to claim 10, wherein the depilation mechanism further comprises a reflecting plate disposed on one side of the emitter away from the cold compress portion.

15. The portable depilation instrument according to claim 14, wherein the reflecting plate is clamped to a side of the heat conductive component away from the first vent.

16. The portable depilation instrument according to claim 1, wherein the first heat dissipation fin is disposed between the one inlet and the fan air inlet.

17. The portable depilation instrument according to claim 1, wherein the heat conductive element is a heat conductive tube.

18. The portable depilation instrument according to claim 1, wherein the first vent is disposed at one end of the heat conductive component facing the fan.

19. The portable depilation instrument according to claim 18, wherein the fan is further provided with a fan air outlet sleeved on the first vent.

20. The portable depilation instrument according to claim 19, wherein a size of the fan air outlet is greater than a size of the first vent.
